

ONE-PAGE SUMMARY REPORT

1. DATA CLEANING CHALLENGES ENCOUNTERED

Initial data preparation resolved critical quality issues to ensure accurate downstream exploratory analysis:

- **Missing Values:** Gaps in CustomerID and Description were dropped or assigned placeholders to prevent distortions in customer-centric metrics and aggregate statistics.
- **Negative Transactions:** Negative values in Quantity and UnitPrice (representing returns/cancellations) were isolated to protect the validity of baseline sales trends.
- **Extreme Outliers:** Massive transaction volumes (orders over 80,000 units) that compressed boxplots were masked using visual filtering (showfliers=False) to reveal core distributions.

2. KEY EXPLORATORY DATA ANALYSIS (EDA) FINDINGS

- **Low-Value Inventory Dominance:** Product prices are heavily right-skewed. Most transactions are clustered tightly between 0 and 3 units, proving that sales volume is driven entirely by affordable items.
- **Geographic Revenue Asymmetry:** A deep comparative analysis across international trade zones showed extreme performance variations, with a single dominant market capturing the vast majority of platform revenue.

3. TOP STRATEGIC INSIGHTS

- **High Market Concentration Risk:** The United Kingdom commands a staggering **89.4%** of total revenue share among top countries. This absolute dependency exposes the business to severe risks from localized economic shocks.
- **Volume-Driven Revenue Architecture:** Profitability relies purely on high purchase volumes of low-cost items rather than high-margin premium sales, making customer acquisition and fast inventory turnover critical.
- **B2B & Wholesale Opportunities:** While Germany and France provide high order frequency, regions like the Netherlands, Australia, and Japan show significantly larger transaction sizes, representing prime targets for bulk B2B marketing.
- **Operational Revenue Leakage:** Substantial negative values in sectors like the USA and Sweden indicate notable product returns or cancellations, highlighting areas that require operational audits.

Strategic Action: Diversify the geographical footprint to reduce UK concentration risk, and implement tailored wholesale price models for bulk-buying regions like the Netherlands and Australia.